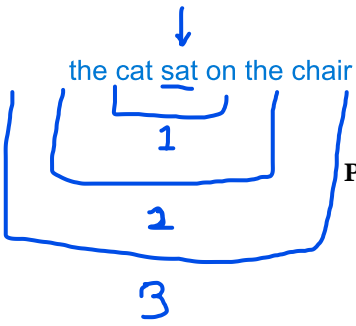


Assignment12 – Deep Learning for NLP

Given: July 8 Due: July 13

small window= syntactic (grammar)

large window= semantic (meaning and topic)

Problem 12.1 (Continuous Bag of Words: Design Trade-Offs)

In the CBOW algorithm, explain the trade-offs regarding the

1. …size of the context window

Solution: A larger context window yields a better embedding because it can take more context knowledge into account. But it also makes the inputs to the network larger, thus requiring more weights and possibly more training time. It may also require a larger hidden layer to allow representing the more fine-grained context knowledge.

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2. …dimension of the hidden layer

Solution: A bigger hidden layer makes it easier to encode all necessary information, and the learned embedding will be more accurate. But it increases the size of the vectors that represent the words, thus making all subsequent operations (for which the word embedding is just preparation) more complex.

(the, cat, on, the, chair)

Problem 12.2 (Back Propagation vs. Feedback)

We compare back propagation and feedback/recurrency in neural networks.

backprop
(
1. prediction error
2. adjust weights
)
- training method
- updates wts
- no memory mechanism

feedback
(output goes directly into input)

- network architecture
- training and interference
- stores/returns info
- provides memory

1. Explain (in about 1 sentence) their commonalities.

Solution: Both use the activation available at the end of a feed-forward network to improve the behavior of the previous layers of the network.

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2. Explain (in about 2 sentences) their differences.

Solution: Back propagation is used at training time to change the weights in the earlier layers. Feedback changes the entire structure of the network, affecting both training and deployment, by making the later activations available as additional inputs to earlier layers.

Problem 12.3 (Context Information in RNNs for NLP)

Using examples, explain the difficulty of taking context into account for NLP and how the following neural network design choices aim at tackling it.

eg. i saw a bat ~ ambigious
eg. sarah spoke to emily bc she was worried ~ who?

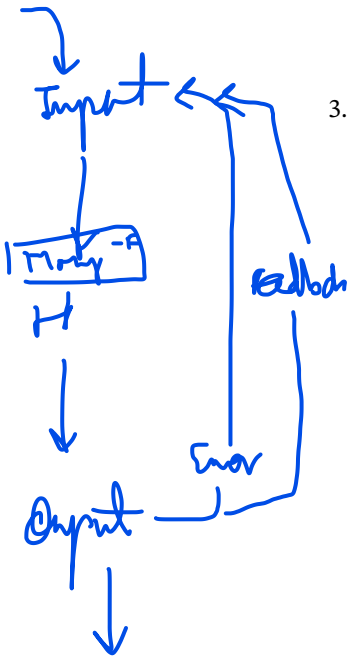
1. Recurrency

Solution: Recurrency allows the network to have memory and thus remember previous context words. Consider the pronoun `him` in the sentence `Eduardo told me that Miguel was very sick so I took him to the hospital.` The preceding words are needed to understand what the pronoun refers to.

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2. Bidirectional RNNs

Solution: Plain recurrency typically only allows the network to remember the *preceding* context words. But NLP tasks often require the *succeeding* context as well. For example, if the above sentence continued `to see him`, the resolution of the first `him` changes. Bidirectional RNNs consider both directions so that they can better use context on both sides.



3. Long short-term memory

Solution: The distance over which context knowledge is relevant can be very large. For example, in the above sentence, the resolvent of the pronoun occurs 11 words before the pronoun. In general, the distance may be much bigger even.
Using such a large context window would be inefficient. LSTM is a network design technique that tries to allow the network to remember more selectively, by using separate cells in charge of forgetting/reinforcing different components of the input and/or the fed-back state. If this is trained well, it allows for remembering the relevant long-distance context while forgetting everything in between.

LSTM
- forget gate
- input gate
- output gate